

1) Best Model Hyperparameters

After tuning the number of epochs, layers, neurons, learning rate and activation functions, the best performing model used a *sigmoid output activation*. It reached a training accuracy of 0.894 (loss 0.279) and a test accuracy of 0.896.

Hyperparameter	Best Value
Hidden layers	1 hidden Dense layer
Neurons in hidden layer	3
Hidden activation	Linear (default)
Output layer	Dense(1)
Output activation	Sigmoid
Loss function	Binary Crossentropy
Optimizer	SGD
Learning rate	0.01
Epochs	100
Batch size	32
Training accuracy	0.894
Training loss	0.279
Test accuracy	0.896

Comparison of all experiments

Experiment	Architecture	Output activation	Optimizer (LR)	Epochs	Train Acc	Train Loss
Base model	Dense(1)	Linear	SGD (0.001)	1	0.667	4.40
More epochs	Dense(1)	Linear	SGD (0.001)	100	0.886	1.70
Extra layer	Dense(1)+Dense(1)	Linear	SGD (0.001)	100	0.839	2.60
More neurons	Dense(3)+Dense(1)	Linear	SGD (0.001)	100	0.871	2.07
Three layers	Dense(3)+Dense(1)+Dense(1)	Linear	SGD (0.001)	100	0.839	2.60
Higher learning rate	Dense(3)+Dense(1)	Linear	SGD (0.01)	100	0.871	2.07
Sigmoid (best)	Dense(3)+Dense(1)	Sigmoid	SGD (0.01)	100	0.894	0.279

Conclusion: adding the sigmoid output activation made the biggest difference, dropping the loss from about 2.0 to 0.28 and giving the highest accuracy.

2) Loss / Accuracy Plot of the Best Model

The curves below show the training loss decreasing and the accuracy increasing by over 100 epochs for the best (sigmoid) model: *accuracy: 0.8941 - loss: 0.2794*

